

Data Cleaning & Reporting Automation

Objective: Automate data cleaning and reporting using Python.

Dataset (sales_data.csv)

```
ID,Name,City,Sales
1,Alice,New York,500
2,Bob,Chicago,700
3,Charlie,,600
4,Alice,New York,500
5,David,Los Angeles,
6,Eva,Chicago,800
7,Frank,New York,900
8,Eva,Chicago,800
```

Python Code

```
import pandas as pd
df=pd.read_csv('sales_data.csv')
df=df.drop_duplicates()
df['City']=df['City'].fillna('Unknown')
df['Sales']=df['Sales'].fillna(df['Sales'].mean())
df.to_csv('cleaned_sales_data.csv',index=False)
summary=df.groupby('City')['Sales'].sum()
with open('report.txt','w') as f:
    f.write(summary.to_string())
```

Expected Output

```
Chicago 1500
Los Angeles 700
New York 1400
Unknown 600
```

Conclusion

Duplicates removed, missing values handled, cleaned dataset and report generated automatically.